



Could Electrical Wave Blaster Provide A New Way To Extinguish Fires?

Recently, researchers have discovered that application of large alternating current (AC)-producing oscillating electric fields can be used to suppress flames rapidly



Electrical wave blaster to extinguish fires (Credit: <http://deskarati.com>)



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Fire is a powerful force! If managed properly, it has many uses; but if left unmanaged, it can be destructive. Fire has the power to support the survival of mankind as well as to destroy everything that comes in contact with it.

In earlier times, ways to extinguish fires were based on use of water and sand; but with technological developments, many other means of extinguishing fire, like use of foam, dry powder, carbon-dioxide and wet chemicals, have come into existence, while others like use of electric fields and sound waves are being developed.

Water- and carbon-dioxide-based fire-extinguishers have become a necessity and an integral part of residential buildings, offices, public infrastructures and factories. However, fire accidents in vehicles, homes, factories, forests and so on are on the rise, resulting in huge losses of life and property.

Development of various types of fire extinguishers to put out fires started with the understanding of different types of sources/fuels of fire—called classes of fire. Risk of fire from its different classes determines which fire extinguisher type is needed—it is important to ensure that the

chosen fire extinguisher is of the right size and weight, and the right kind.

Presently, there is no single extinguisher type that works well on all classes of fire. Fire accidents, in general, and in enclosed environments, in particular, such as houses, airplanes, vehicles and ships, continue to be a major cause of material destruction and loss of life.

Scientists are exploring novel fire-suppressing approaches based on physical techniques, as opposed to current methods that attack the chemistry of the flame by suppressing it. By understanding the electromagnetic and acoustic nature of fire, they are working to develop two different schemes for flame suppression—one that involves application of electric fields and the second that uses sound.

Currently, using a handheld electrode, scientists have been able to suppress small methane gas and liquid fuel fires. This method creates an oscillating field that induces a rapid series of waves, which displace the combustion zone from the fuel source. In other words, the electric field creates an ionic wind in the flame, which helps in blowing it out.

In another experiment, the flame was extinguished using an acoustic field gener-